12. The term adiabatic change of temperature stands for:
    (a) Heating of ascending air.
    (b) Heating of descending air.
    (c) Cooling of ascending air.
    (d) Heating or cooling of ascending or descending air through expansion or compression.
13. Consider the following about the Daylight Saving Time and select the correct answer using the code given below:
    1. Time is set ahead one hour in the spring and is set back one hour in the autumn in the Northern Hemisphere.
    2. Time is set ahead on the first Sunday in April and is set back on the last Sunday in October.
    3. This adjustment in time is done in all the states of USA.

    Codes:

    (a) 1 and 2 are correct
    (b) 1 and 3 are correct
    (c) 2 and 3 are correct
    (d) 1, 2 and 3 are correct
14. Consider the following statements about the tropical cyclones and select the correct answer using the code given below:
    1. Tropical cyclones have closed isobars.
    2. The isobars are circular in shape.
    3. They give human friendly rainfall.
    4. They have an eye in the centre.

    Codes:

    (a) 1, 2 and 3 are correct
    (b) 1, 2 and 4 are correct
    (c) 1, 3 and 4 are correct
    (d) 1, 2, 3 and 4 are correct
15. Consider the following statements about the tropical cyclones and select the correct answer using the code that follows:
    1. In tropical cyclones, air is essentially homogenous.
    2. Tropical cyclones are without fronts.
    3. The cyclonic motion in tropical cyclones begins because of coriolis force.
    4. Tropical cyclones are destructive by nature.

    Codes: